

1. You are given an array `people` where `people[i]` is the weight of the *i*th person, and an infinite number of boats where each boat can carry a maximum weight of `limit`. Each boat carries at most two people at the same time, provided the sum of the weight of those people is at most `limit`. Find the minimum number of boats to carry every given person.

Example 1:

Input: `people = [1,2]`, `limit = 3`

Output: 1

Explanation: 1 boat (1, 2)

Example 2:

Input: `people = [3,2,2,1]`, `limit = 3`

Output: 3

Example 3:

Input: `people = [3,5,3,4]`, `limit = 5`

Output: 4

2. Given an integer array `nums`, return the maximum possible sum of elements of the array such that it is divisible by three.

Example 1:

Input: `nums = [3,6,5,1,8]`

Output: 18

Explanation: Pick numbers 3, 6, 1 and 8 their sum is 18 (maximum sum divisible by 3).

Example 2:

Input: `nums = [4]`

Output: 0

Example 3:

Input: `nums = [1,2,3,4,4]`

Output: 12

3. Given an *n* x *n* binary matrix `image`, flip the image horizontally, then invert it, and return the resulting image. To flip an image horizontally means that each row of the image is reversed. For example, flipping `[1,1,0]` horizontally results in `[0,1,1]`. To invert an image means that each 0 is replaced by 1, and each 1 is replaced by 0. For example, inverting `[0,1,1]` results in `[1,0,0]`.

Example 1:

Input: `image = [[1,1,0],[1,0,1],[0,0,0]]`

Output: `[[1,0,0],[0,1,0],[1,1,1]]`

Explanation: First reverse each row: `[[0,1,1],[1,0,1],[0,0,0]]`.

Then, invert the image: `[[1,0,0],[0,1,0],[1,1,1]]`

Example 2:

Input: `image = [[1,1,0,0],[1,0,0,1],[0,1,1,1],[1,0,1,0]]`

Output: `[[1,1,0,0],[0,1,1,0],[0,0,0,1],[1,0,1,0]]`

4. You are given a large integer represented as an integer array `digits`, where each `digits[i]` is the *i*th digit of the integer. The digits are ordered from most significant to least significant in left-to-right order. The large integer does not contain any leading 0's. Increment the large integer by one and return the resulting array of digits.

Example 1:

Input: digits = [1,2,3]

Output: [1,2,4]

Explanation: The array represents the integer 123.

Incrementing by one gives $123 + 1 = 124$.

Thus, the result should be [1,2,4].

Example 2:

Input: digits = [4,3,2,1]

Output: [4,3,2,2]

Example 3:

Input: digits = [9]

Output: [1,0]

5. Given an array of strings words (without duplicates), return all the concatenated words in the given list of words. A concatenated word is defined as a string that is comprised entirely of at least two shorter words (not necessarily distinct) in the given array.

Example 1:

Input: words = ["cat","cats","catsdogcats","dog","dogcatsdog","hippopotamuses","rat","ratcatdogcat"]

Output: ["catsdogcats","dogcatsdog","ratcatdogcat"]

Explanation: "catsdogcats" can be concatenated by "cats", "dog" and "cats";

"dogcatsdog" can be concatenated by "dog", "cats" and "dog";

"ratcatdogcat" can be concatenated by "rat", "cat", "dog" and "cat".

Example 2:

Input: words = ["cat","dog","catdog"]

Output: ["catdog"]